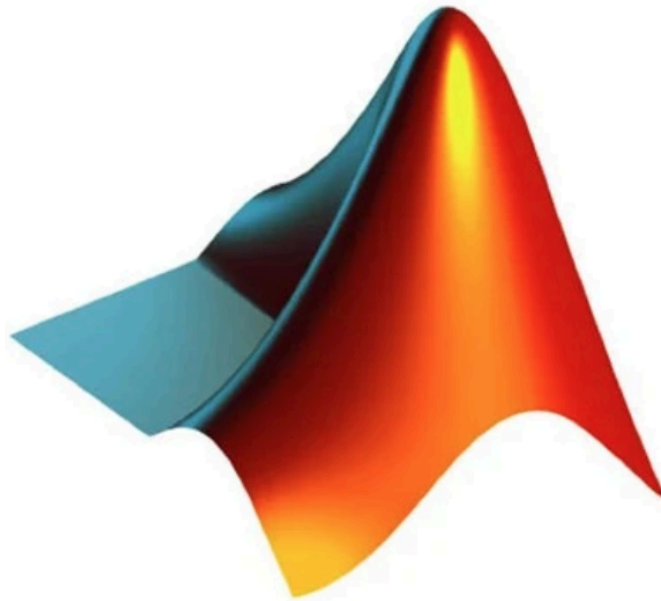


MATH 284 MATLAB Project #3 - Vector Space Explorer



MATLAB

$\mathbb{R}^3$

set of real vectors with three components

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \vec{a} + \vec{b} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{bmatrix}$$

- 1) given $\vec{a} \in V$ and scalar c , then $c\vec{a} \in V$ ✓
- ➡ 2) given $\vec{a} \in V$ and $\vec{b} \in V$, then $\vec{a} + \vec{b} \in V$ ✓

Project Objective

Develop an interactive MATLAB application that allows users to investigate vector spaces, determine whether sets form subspaces, compute bases, visualize coordinate systems, and explore how linear transformations affect vector spaces.

Rather than simply solving problems, this application lets users explore the structure of vector spaces.

Concepts And Textbook Chapters/Sections Covered

Chapter 4: Vector Spaces

- 4.1 Vector Spaces and Subspaces
 - 4.2 Null Spaces, Column Spaces, Row Spaces, and Linear Transformations
 - 4.3 Linearly Independent Sets, Bases
 - 4.4 Coordinate Systems
 - 4.5 The Dimension of a Vector Space
-
- Vector spaces
 - Subspaces
 - Null spaces
 - Column spaces
 - Row spaces
 - Bases
 - Linear independence
 - Coordinate systems
 - Dimension
 - Linear transformations

GUI Layout

Vector Space Explorer	
Input	
Matrix A	
[	]
Vector Set	
[	]
Analysis Tools	
[Subspace Test]	
[Basis Finder]	
[Coordinate Converter]	
[Transformation Viewer]	
Space Properties	
Basis	
Dimension	
Rank	
Independent?	
Dependent?	
Visualization	
Vector Plot	
Basis Vectors	
Column Space	
Null Space	
Advanced	
[Column Space]	
[Row Space]	
[Null Space]	
[Coordinate Systems]	

Necessary MATLAB Files For This Project

VectorSpaceExplorerApp.m

Responsibilities:

- *GUI*
- *User interaction*
- *Display plots*
- *Display explanations*

VectorSpace.m

Stores:

- vectors
- basis
- dimension
- subspace
- rank

BasisAnalyzer.m

Implements:

- *Basis computation*
- *Minimal spanning sets*
- *Linear independence*

CoordinateSystemConverter.m

Converts standard coordinates to coordinates relative to the basis

SubspaceAnalyzer.m

Checks the following conditions:

- *Closure under addition*
- *Closure under scalar multiplication*
- *Contains zero vector*

And computes:

- *Column Space*
- *Row Space*
- *Null Space*

TransformationExplorer.m

Input:

- *Transformation Matrix*

Outputs:

- *Original basis*
- *Transformed basis*
- *Transformed vectors*

VectorSpacePlotter.m

Responsible for:

- *Basis vectors*
- *Span visualization*
- *Coordinate grids*
- *Subspaces*

Necessary MATLAB Methods For This Project

createComponents()

analyzeSpace()

updatePlots()

clearGUI()

findBasis()

calculateDimension()

isSubspace()

span()

computeBasis()

removeDependentVectors()

verifyBasis()

convertToBasis()

convertToStandard()

verifyCoordinates()

columnSpace()

rowSpace()

nullSpace()

testSubspace()

applyTransformation()

animateTransformation()

plotTransformation()

plotBasis()

plotSpan()

plotCoordinateSystem()

plotSubspace()

GUI Functions

- Subspace tester
- Space explorer
- Basis finder
- Coordinate converter
- Dimension analyzer

Pseudocode

START PROGRAM

Create GUI

Initialize

Matrix input

Vector input

Plot area

Results panel

WHEN Analyze pressed

Read vectors

Determine

Basis

Dimension

Rank

Linear independence

Display results

WHEN Subspace selected

Compute

Column Space

Row Space

Null Space

Display basis

Plot space

WHEN Coordinate Converter selected

Read basis

Read vector

Convert coordinates

Display conversion

WHEN Transformation Viewer selected

Read transformation matrix

Apply transformation

Plot before

Plot after

WHEN Clear selected

Reset GUI

END PROGRAM

$$\begin{bmatrix} 0 & 1 & 8 & 9 \\ 1 & 9 & 7 & 3 \\ 9 & 8 & 2 & 5 \\ 1 & 7 & 3 & 8 \end{bmatrix}$$

Row A

Col A

Nul A

$$\begin{bmatrix} 5 & 8 & 1 & 0 \\ 2 & 3 & 5 & 0 \\ 9 & 8 & 0 & 0 \\ 6 & 7 & 2 & 4 \end{bmatrix}$$

linear algebra

$\lambda_1 = 3 \quad \lambda_2 = 7 \quad \det(A) = 0$